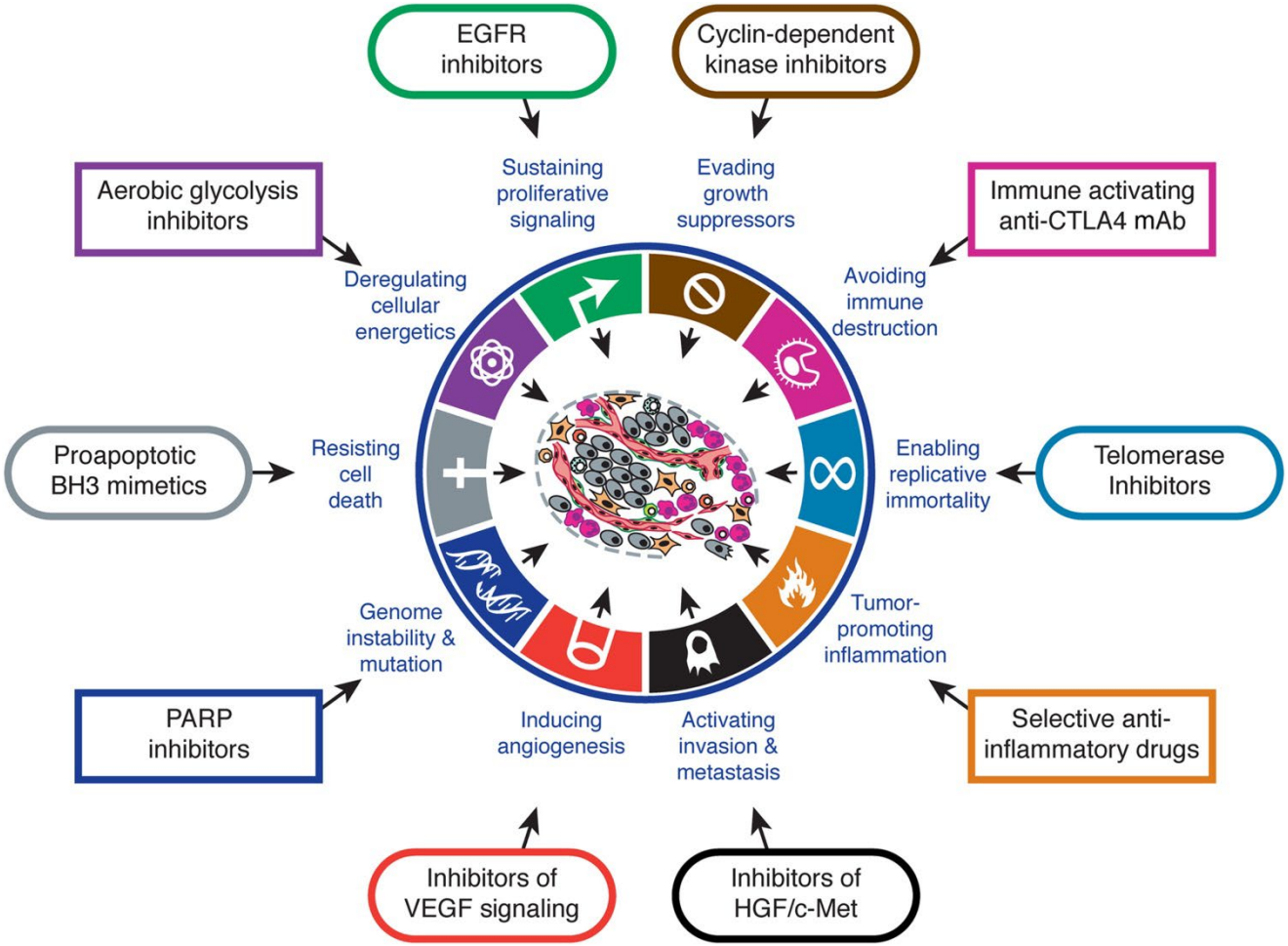
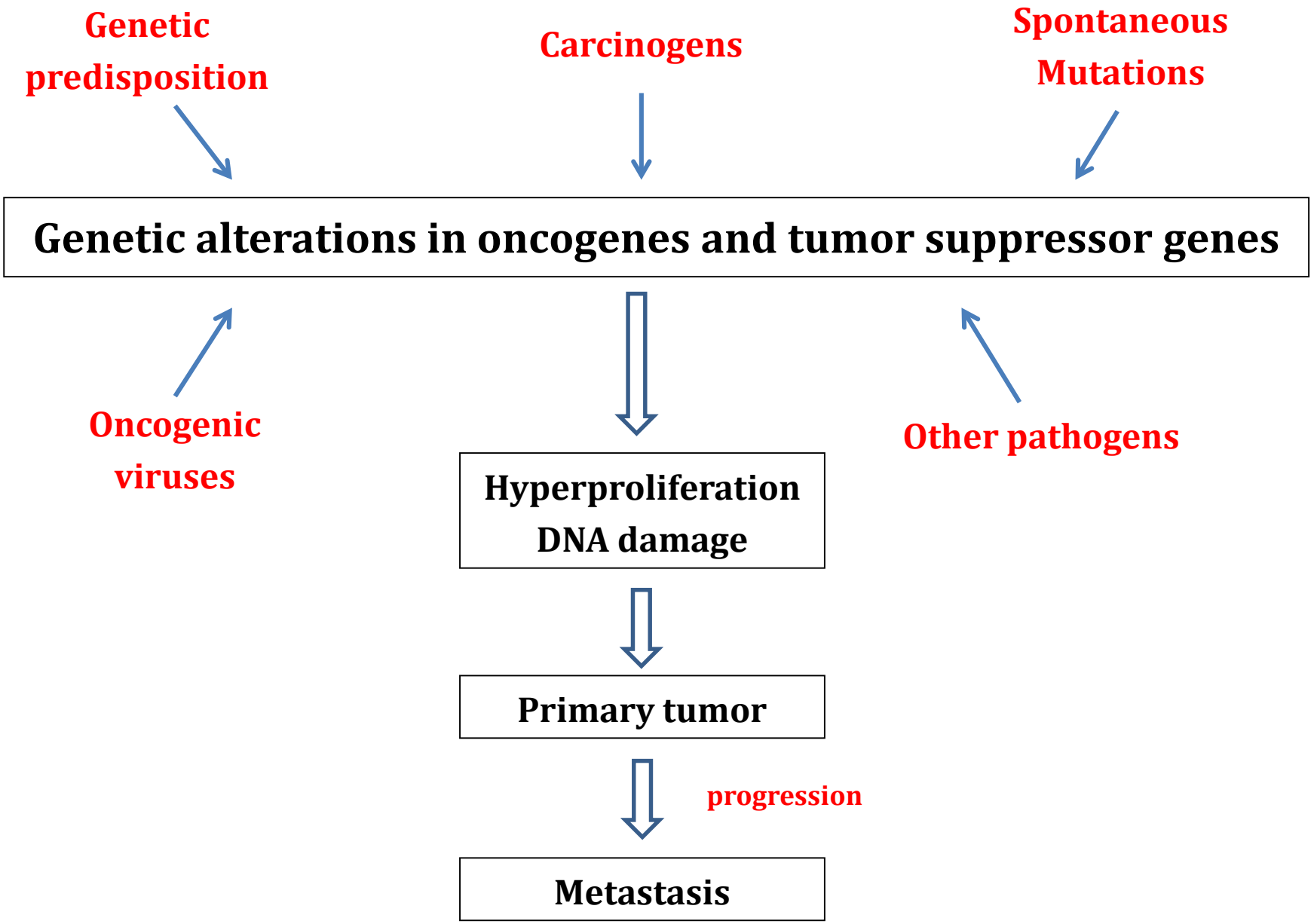
The hallmarks of cancer

A complex and multistep disease: genetic and environmental factors




Genetic predisposition
Carcinogens
Spontaneous Mutations
Genetic alterations in oncogenes and tumor suppressor genes
Oncogenic viruses
Other pathogens
Hyperproliferation
DNA damage
Primary tumor
progression
Metastasis

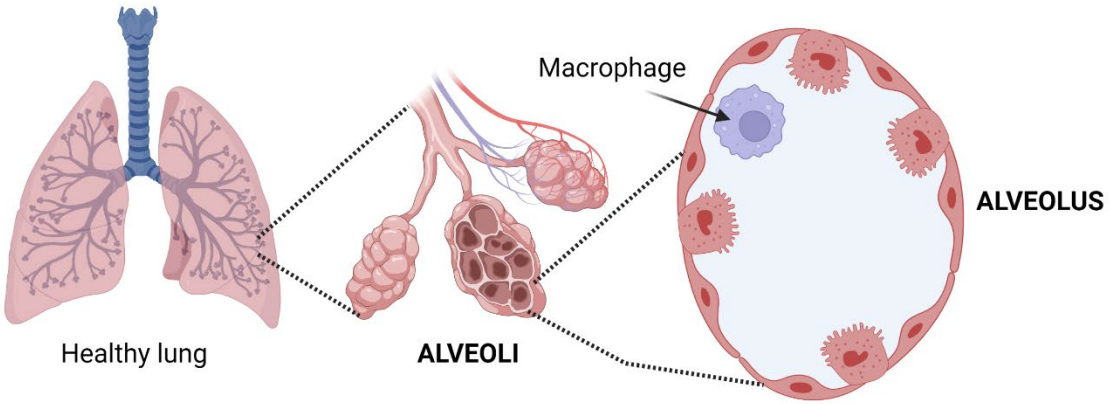
Chemical carcinogens and cancer

Effects of tobacco smoke

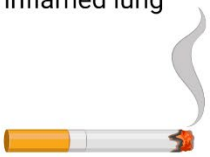
TOBACCO SMOKE:
4500 chemical compounds
100 carcinogenic

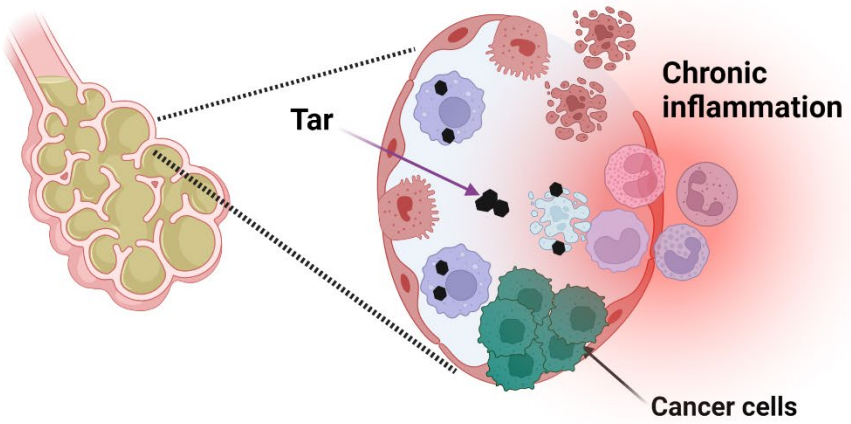
- DNA damage: activation of oncogenes, inactivation of TSG
- Chronic inflammation





TOBACCO:
Continuous DNA damage: **cancer**
Chronically inflamed lung

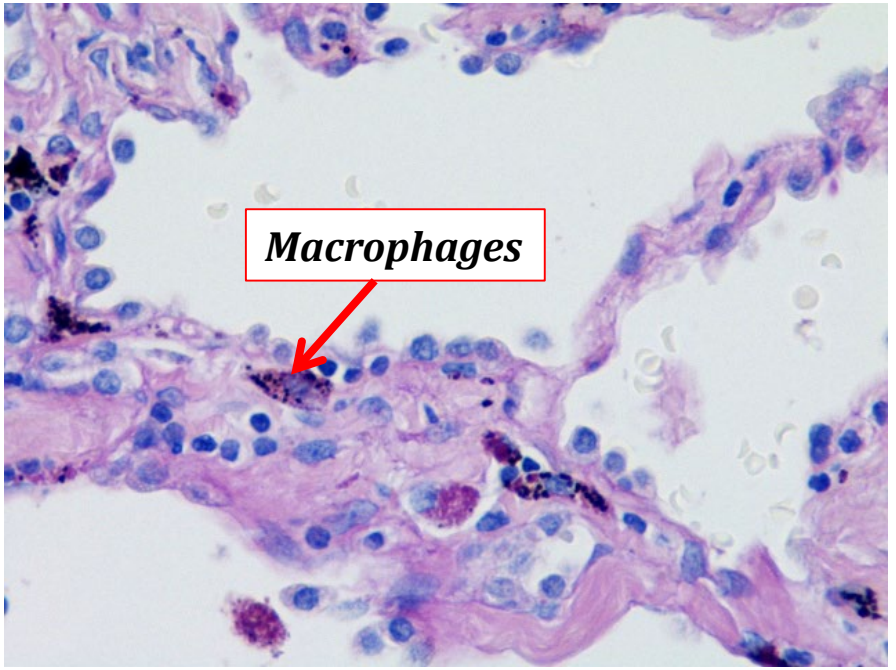




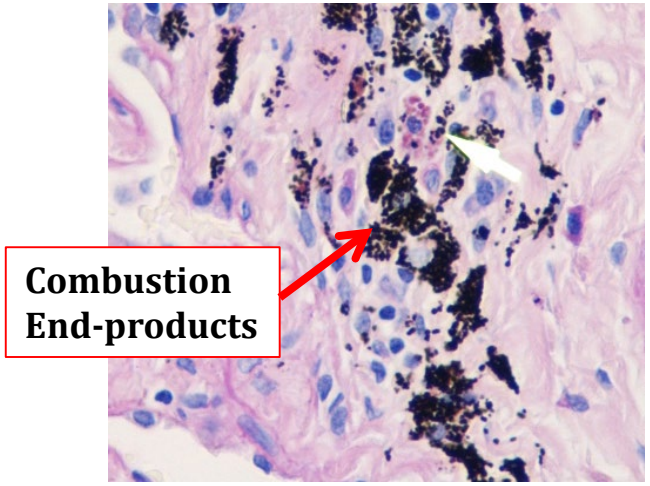
healthy lung



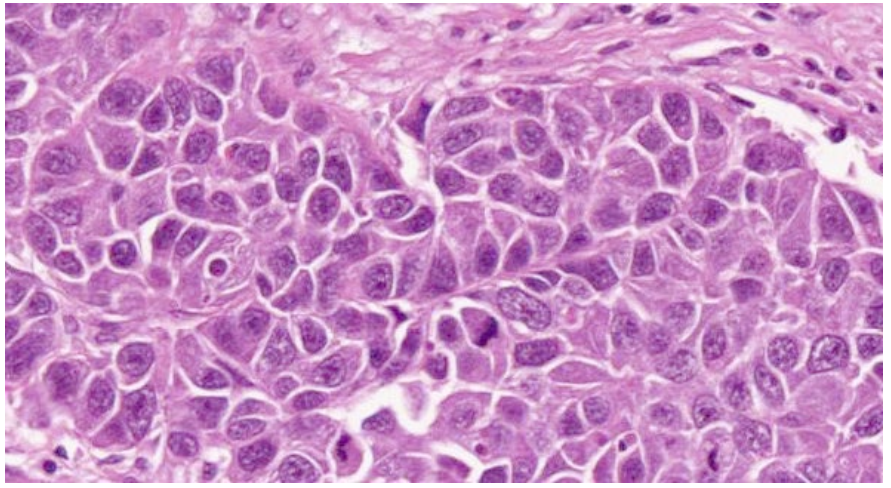
Lung from a heavy smoker



Lung from a heavy smoker



Lung cancer



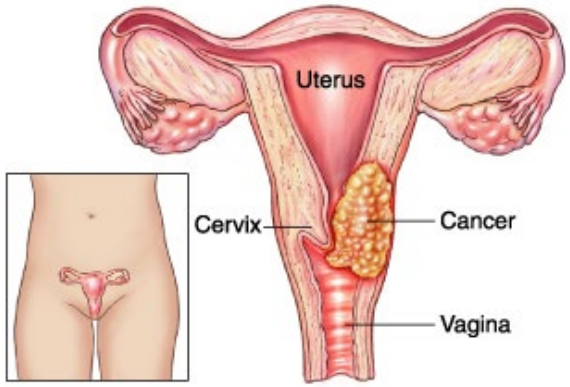
Slide that you have to study

Oncogenic viruses: *The case of papillomavirus*

Viral proteins

E7

E6



p53

P

Rb

Inactivation

E2F

Excessive proliferation

Cells unprotected from DNA damage

Cancer: accumulation of sequential genetic alterations

Familiar adenomatous polyposis

Germline mutations in the *APC* (adenomatous polyposis coli) gene

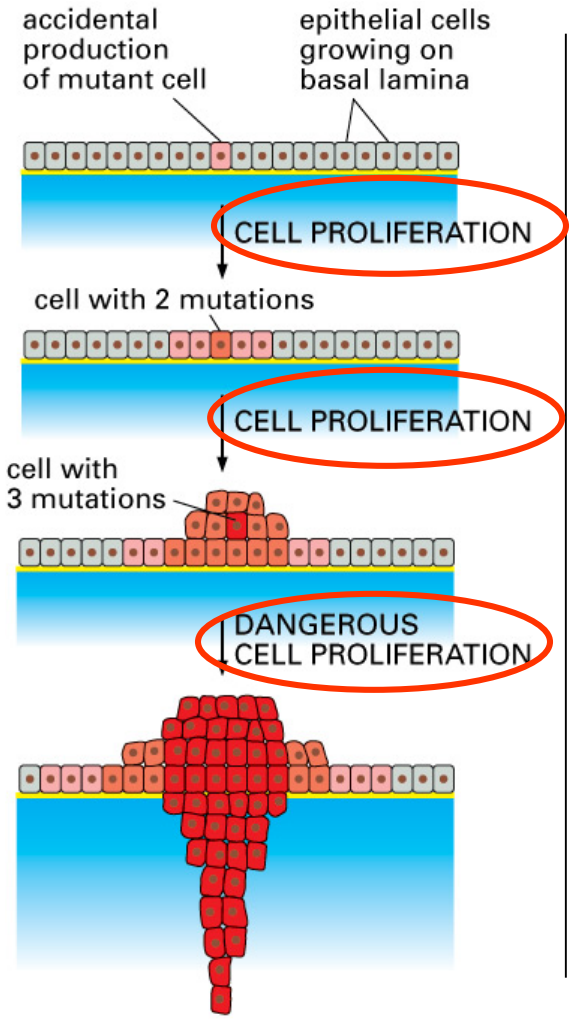


Figure 23–11. Molecular Biology of the Cell, 4th Edition.

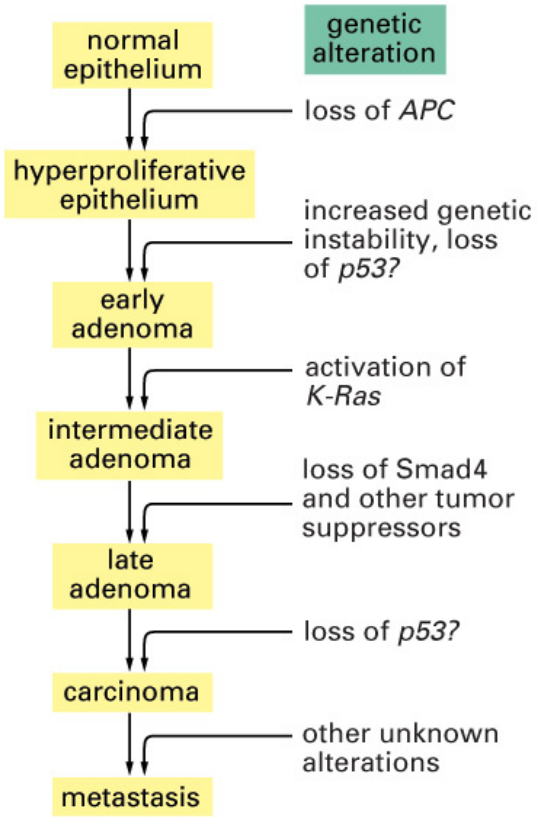


Figure 23–41. Molecular Biology of the Cell, 4th Edition.

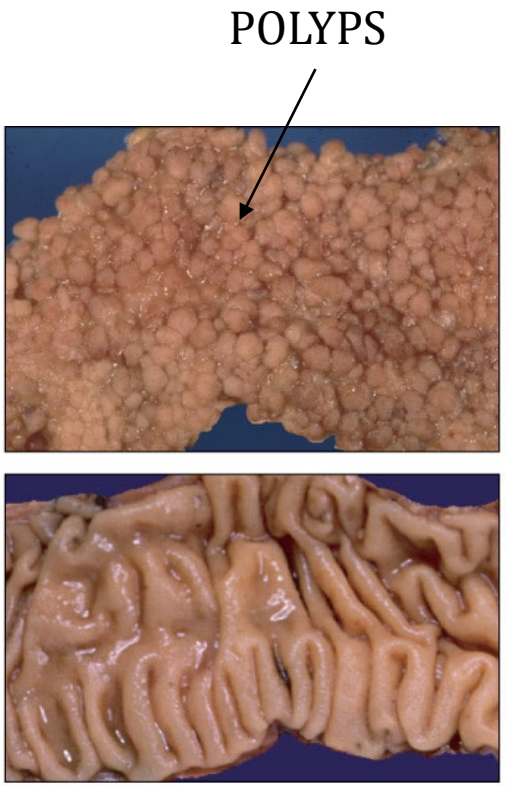


Figure 23–39. Molecular Biology of the Cell, 4th Edition.

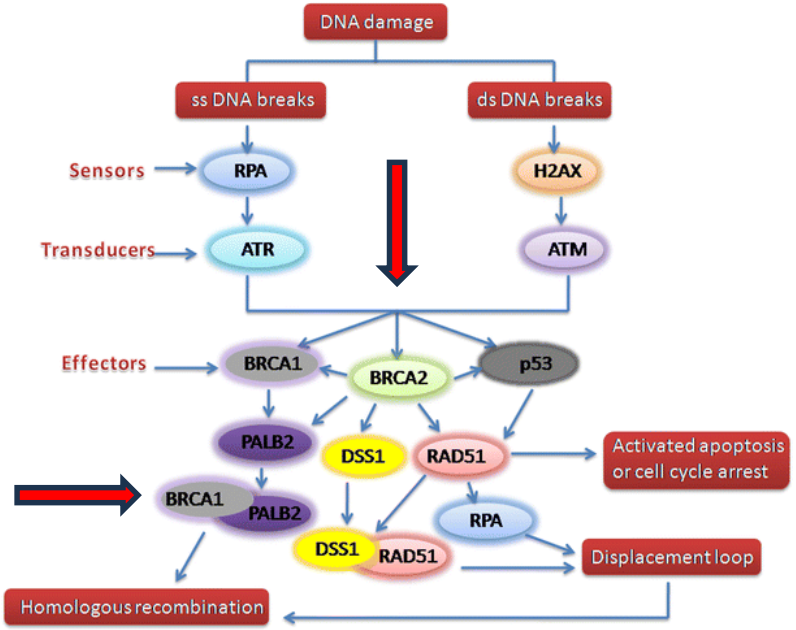
Cancer progression: ↑ Genomic instability

Karyotype of a breast cancer cell taken from a metastatic tumor

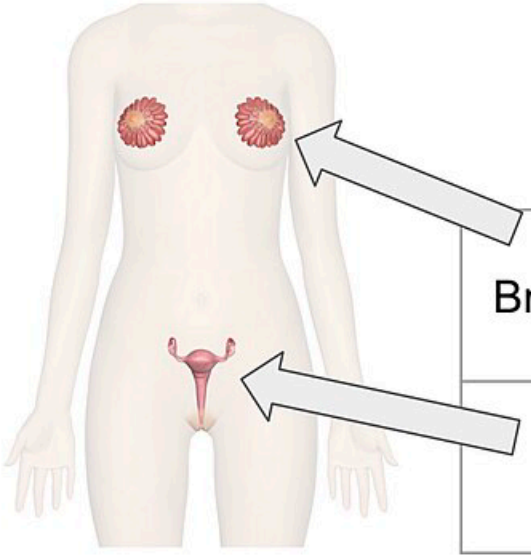


BRCA1/2 mutations and genomic instability





Hereditary Breast and Ovarian Cancer Syndrome (HBOC)



	No BRCA1 or BRCA2 mutations	With BRCA1 mutations	With BRCA2 mutations
Breast cancer	12%	55-72%	45-69%
Ovarian cancer	1-2%	39-44%	11-17%

Clinical scenario to discuss



-A 29-year-old woman with no personal history of cancer.
Reason for consultation: Genetic counseling due to a striking family history of breast and ovarian cancer:

-Mother: diagnosed with breast cancer at age 43, died at 46.
-Maternal aunt: Diagnosed with ovarian cancer at age 50, died at 52.
-Maternal grandmother: Breast cancer at age 60, died at 65.
-Father: Healthy, no personal or family history of malignancy.

After a **multigene hereditary cancer panel: BRCA1 pathogenic variant** detected (frameshift/truncating mutation), confirming **Hereditary Breast and Ovarian Cancer Syndrome (HBOC)**.

"I feel perfectly healthy, but since my mom and aunt both died young, and now I carry the same BRCA1 mutation, what should I do next?"

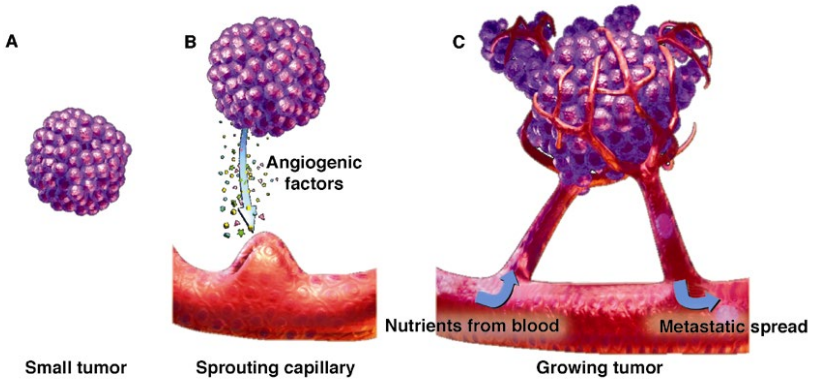
Slide that you have to study (excluding the video)

Tumor angiogenesis and metastasis

-Without blood vessels: 1-2 mm^3

-Tumor angiogenesis (recruitment of vessels)

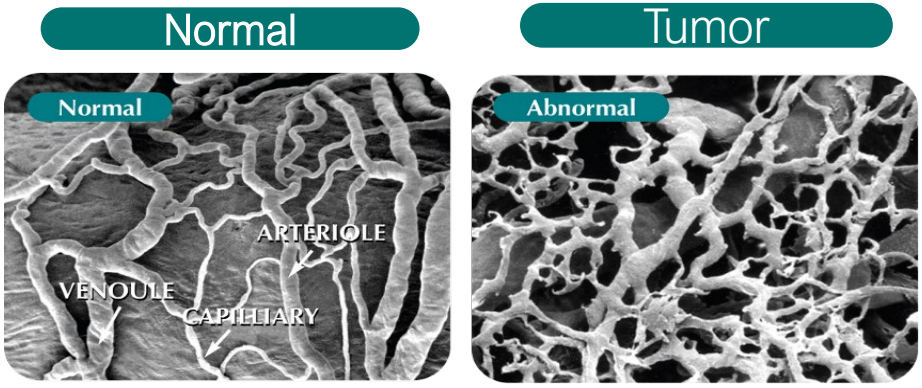
- Tumor growth
- Cell escape (bloodstream)



Video antiangiogenesis

https://www.youtube.com/watch?v=Ep_nCSEDeAE

(disorganized, ↑ VEGF)





Video Escape from the immune system

https://www.youtube.com/results?search_query=immunotherapy+the+path+to+cure+cancer

Mary Brunkow, Fred Ramsdell and Shimon Sakaguchi are awarded the Nobel Prize in Physiology or Medicine 2025 for their fundamental discoveries relating to peripheral immune tolerance. The laureates identified the immune system's security guards, *regulatory T cells*, which prevent immune cells from attacking our own body.

Tumors induce the development of T regulatory cells, which block the effects of immune cells that fight cancer.



Focus on breast cancer

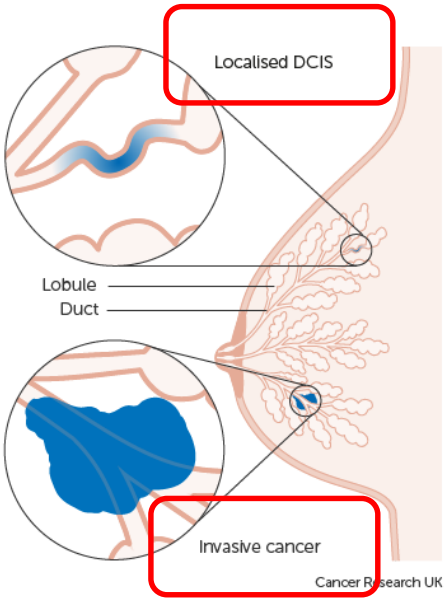
Epidemiological data in Spain

-About 35,000 new cases every year. **First in incidence** among women.

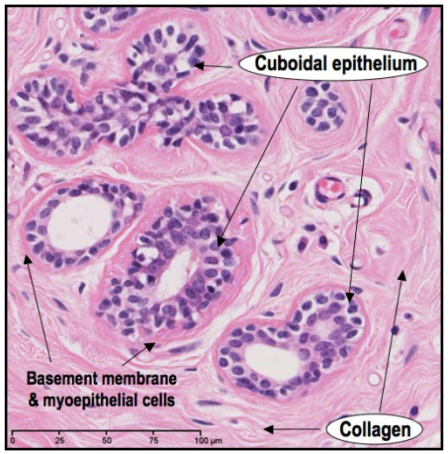
-**Survival** rate 5 years after diagnosis: **>90%.**

-**1 out 8 women** in Spain will develop breast cancer over the course of their lives.

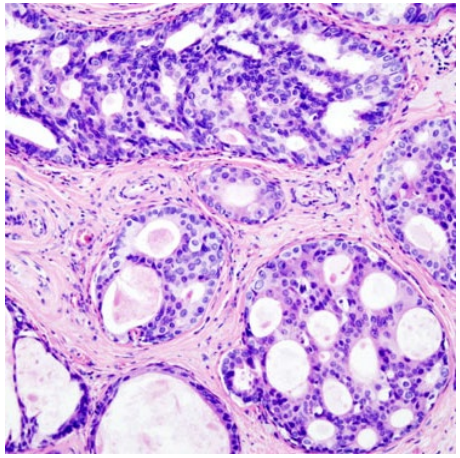
If detected early, possibility of cure is ~100%



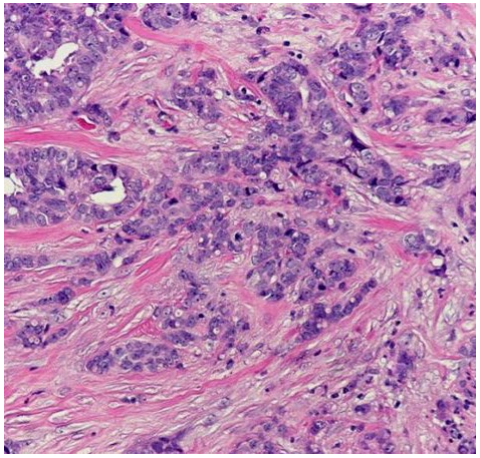
Normal breast



Ductal carcinoma in situ (DCIS)



Invasive carcinoma



Clinical case

Focus on anti-Her-2 therapy

International Journal of Surgery Case Reports 52 (2018) 125–131

Contents lists available at ScienceDirect

International Journal of Surgery Case Reports

journal homepage: www.casereports.com

Case Series

Patients who achieved long-term clinical complete response and subsequently terminated multidisciplinary and anti-HER2 therapy for metastatic breast cancer: A case series

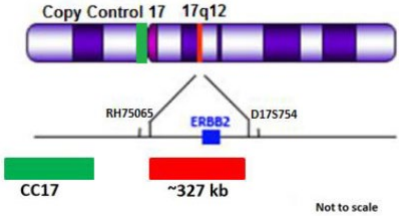
Haruko Takuwa*, Wakako Tsuji, Fumiaki Yotsumoto

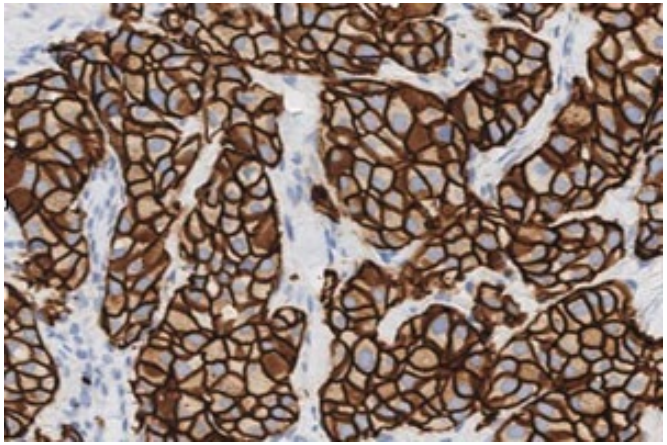
Department of Breast Surgery, Shiga General Hospital, 5-4-30, Moriyama, Moriyama-city, Shiga, 524-8524, Japan

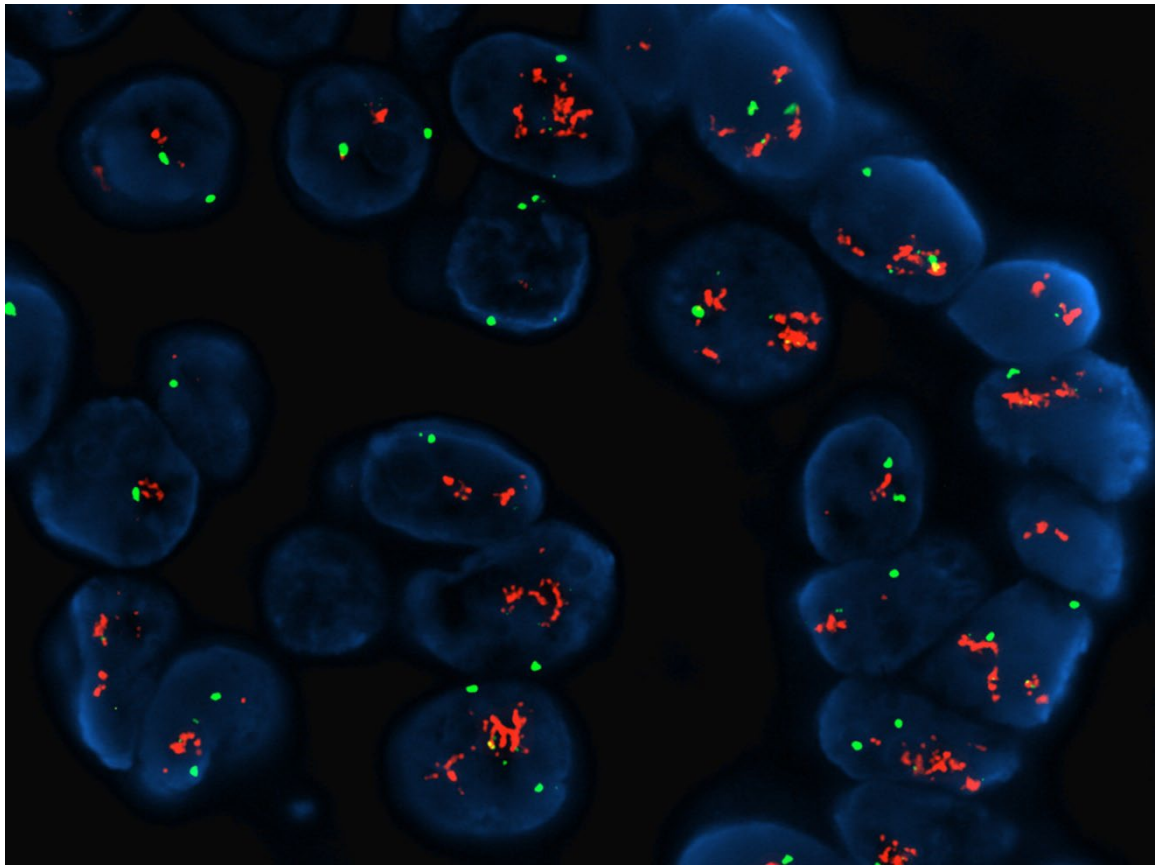
Gene amplification of Her-2 in BREAST CANCER

Her-2: red

Centromere chromosome 17: green



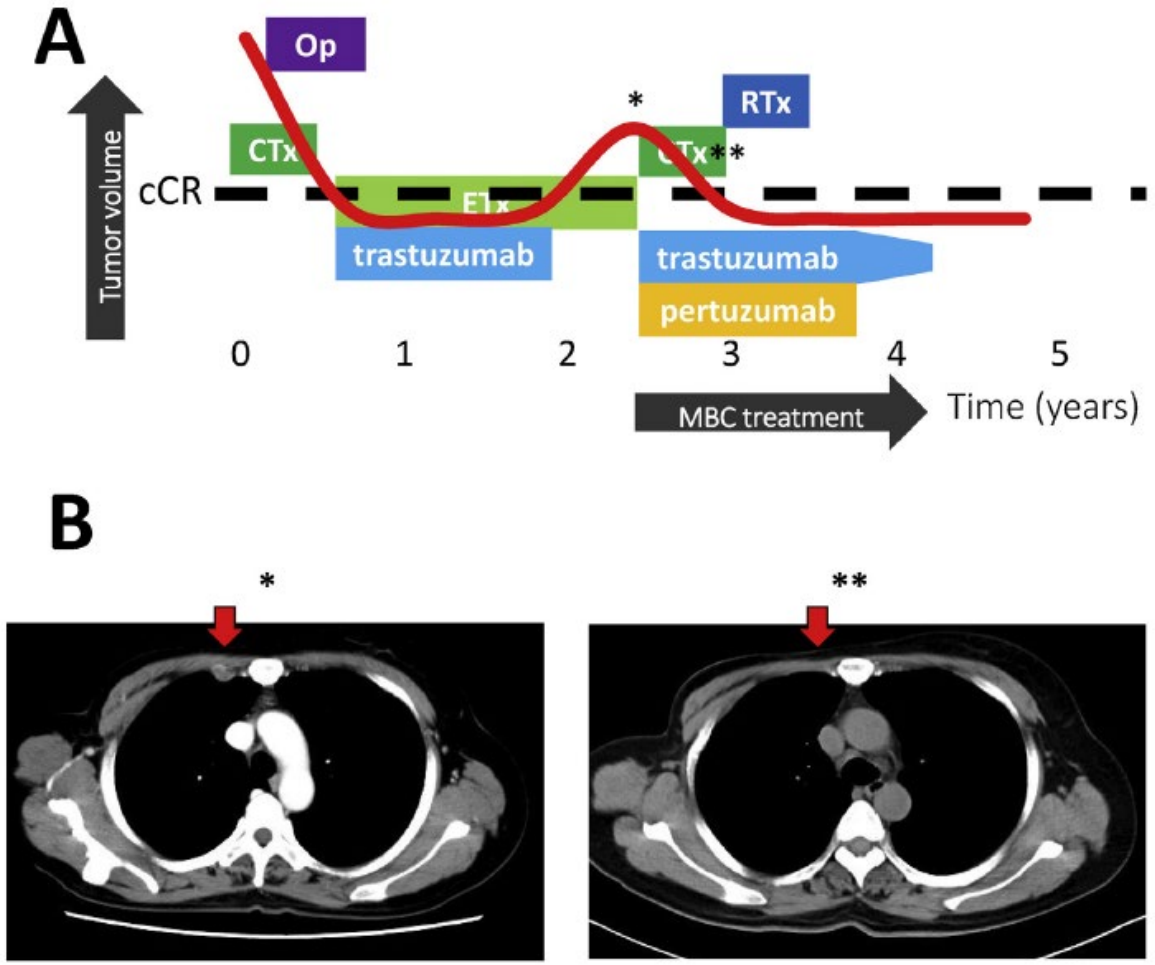




A 41-year-old woman with T3N2M0 disease (stage IIIA, HER2 positive) underwent preoperative chemotherapy using 2 cycles of epirubicin plus cyclophosphamide followed by 2cycles of weekly paclitaxel in 2013.

Surgery with mastectomy and axillary lymph node dissection was then applied and pathological response to chemotherapy was found not to be complete. The patient subsequently received trastuzumab and tamoxifen, but did not undergo post-mastectomy radiotherapy.

At 2 years after surgery, and during treatment, lung and internal mammary lymph nodes **metastases** were found. Thus, treatment using paclitaxel with pertuzumab and trastuzumab was started. After 4 cycles, the patient achieved a complete radiological response. She subsequently underwent irradiation to the chest wall and internal mammary lymph node region, and received maintenance therapy using pertuzumab plus trastuzumab for approximately 18 months. She stopped maintenance therapy at October, 2017. The last follow up was August, 2018 and she has survived for 10 months after termination of anti-HER2 therapy (Figure in the next slide).



Op = operation; **RTx** = radiotherapy; **CTx** = chemotherapy; **ETx:** endocrine therapy; **cCR** = clinical complete response; **MBC** = metastatic breast cancer